

Institute of Actuaries of India

Subject SA3 – General Insurance

May 2023 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

i)

a) As per reinsurance Regulations, every Indian Insurer shall:

- ensure that its Re-insurance arrangements in respect of catastrophe accumulations are adequate;
- have the catastrophe modeling report and the basis along with return period estimates, on which the quantum of catastrophe protection is purchased for each of the perils for the forthcoming financial year duly approved by its Board;
- file a synopsis of the report along-with the Re-insurance Programme.

[2]

b) Higher Reinsurance Cost**Projected Business (Risk profile)**

- Higher business Volume leading to increased exposure
- Higher exposure in earthquake prone areas leading to increase in expected costs
- Higher exposure in areas with older constructions which are more susceptible to earthquakes
- Higher exposure in low lying areas / flood plains leading to increase in expected costs
- Expected increase in optional cover take up like STFI and coverage for earthquake

Change in Reinsurance Program

- Higher Net Retention after Proportional arrangements
 - Lower quota share
 - Lower surplus ceding due to increase in Retention level
 - More risks ceded to treaty (and hence lower expected facultative usage compared to existing norms)
 - Introduction of event limit in treaty (which needs to be covered under non proportional cover)
- Change in Non Proportional Structure
 - Reduction in retention (attachment point)
 - Resulting in higher cost for first layer
 - Increase in Top Layer
 - In line with higher business volumes
 - In line with higher net retention after proportional arrangements
 - In line with wider coverage including other than property risks
- Change in Reinsurance panel
 - Move to reinsurers with higher credit rating and hence higher cost

Other economic factors

- Increase in interest rates leading to higher cost of capital and hence reinsurers requiring higher return on Capital
- Higher cost of materials leading to increase in expected reconstruction costs
- Expected shortage of materials / labour leading to further increase in expected costs (Demand Surge)
- Further strain due to norms / guidelines to settle such claims quickly (with relaxation in usual requirements)

Policy features

- Wider policy coverage – e.g. New for Old (instead of Market value) leading to increase in expected loss costs
- Auto escalation of SI over policy durations leading to increase in expected loss costs

CAT Frequency

- Higher losses experienced by reinsurers leading to increase in RI costs
- Expectation of more CAT events due to climate change leading to increase in RI cost

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ii)

a) Role of CRO in PMC

- Integrate the risks arising out of the proposed product into the company's risk management framework
- Coordinate with Appointed Actuary, Chief Underwriter and other product stakeholders in the company to
 - identify and assess non insurance and residual risks
 - quantify these risks
 - recommend an effective mechanism to minimize these risks to the PMC

[2]

b) Challenges associated with Bharat Griha Raksha Product

Pricing (being Long tenure product with term up to 10 years)

- Difficult to predict future claims : claims inflation – cost of labour, cost of building materials – cement, steel, sand, bricks, glass
- Difficult to predict future reinsurance cost – XL cover may be required to be arranged during each future year with unknown Rate on Line
- Current and future cost of capital / return required by shareholders – opportunity cost
- Investment income – future inflation, interest rates
- Price locked in now and cannot be altered
- Difficult to predict operational cost – employee cost, rentals, IT systems

Product construct / Policy terms and conditions

- No under insurance - to result in higher claim pay out
- No deductible - to result in higher claim pay out
- Covers structure as well as contents on reinstatement / replacement value basis
- Significant premium will go towards terrorism Pool (and may not be required by the policyholder)

Reserving (/IFRS)

- Difficult to predict
 - Cashflows especially with respect to future CAT events (if discounting of reserves is allowed)
 - Attitude towards claiming / expected litigation
 - Increase in claim frequency and severity (to impact segment level / overall PDR requirement, formulating CSM engine)
 - with ageing of buildings
 - climate change leading to increase in cyclones, floods (with associated demand surge)

Impact on Profitability / return on capital

- Upfront commission on entire premium (if acquisition costs cannot be deferred)
- Profit recognition over time along with earning (assuming 1/365th method is used)
- Upfront capital requirement on premium recognition as per existing capital requirement / solvency framework
- May prompt need to have proportional quota share reinsurance to
 - Reduce capital requirement
 - Get upfront commission

Reinsurance

- non proportional reinsurance – availability and cost
 - No of reinstatements, cost of reinstatement
 - Cost of reinsurance / Rate on line by layers
 - Accumulation of risks and possible increase in Top layer over time
- Proportional reinsurance
 - Overall cost of the product will further increase with increase in cost of reinsurance
 - This can impact competitiveness and expected volume growth of the product
- Risk appetite of insurer (and change over time)
 - This may change depending upon underlying profitability

- Growth achieved by insurer
- Access to / availability of capital

Investment and Asset Liability Management

- Liquidity requirement
 - to meet claims from CAT events (with unknown future level of CAT retention, if any),
 - Future policy cancellations requiring refunds
 - Solvency Ratio of insurer (current and projected)
 - How big is this portfolio going to be compared to overall portfolio
 - Cash call facility from proportional reinsurer, if any
- Expected Interest rates – role of central banks, state of economy, target inflation, target GDP growth
- Allowable mismatch between assets and liabilities (threshold at business segment level or overall)
 - E.g. Max gap in DMT allowed by ALCO / Risk Management Committee
 - Reinvestment risk if interest rates reduce over time (but bond price will increase)
 - Short term Vs Long Term – term structure of interest rates
 - Equity / MF investments – can provide liquidity and higher returns but higher market risk
 - Govt securities – safe, liquid, but may offer lower rates
 - Other Private Company Bonds – may provide higher returns but lack liquidity and with higher credit risk
 - Expected future business growth – which can otherwise provide liquidity

[8]

iii)

Share capital

- Permanent form of capital
- Will result in lower return on capital as profit will be spread over larger base
- Will help achieve business growth and help manage solvency simultaneously.
- Need to be compliant with all applicable rules – IRDAI, RBI, SEBI, FDI, FEMA

Price discovery through strategic Stake sell / IPO / FPO

- Insurer may not get the expected price
- Time consuming / long process
- May need to give Board seat to investor with say 10% + stake

Capital infusion by promoter/s

- Faster way to have capital when compared to raising through market.
- Existing shareholders can maintain their existing shareholding and need not dilute their stake

(Max 2)

Other forms of Capital (OFC)

- Preference Shares / Subordinated debt
- Non-convertible, fully paid up and unsecured
- Need to be compliant with all IRDAI, RBI, SEBI, FEMA, FDI rules
- Limits on absolute amount of other forms of capital that can be raised. Insurer can raise lower of
 - 50% of net worth
 - 50% of paid up equity capital including securities premium
- Will boost solvency initially (minimum for 5 years with minimum tenure of 10 years)
- Solvency credit will reduce by 20% each over last 5 years to maturity
- Will help reduce Tax liability
- Gearing effect
- Can have call option after 5 years
- Can exercise call option after 5 years without prior approval of IRDAI if Solvency remains above 180% (20% in excess of 150%) after exercising the call option

- Prior approval of IRDAI required to pay dividend on preference share / interest on subordinated debt if Solvency is /to go below 150%, to result in net loss / increase net loss
- May help achieve business growth and solvency both

(Max 4)

Reinsurance as a source of capital

- Can take advantage of existing solvency norms with additional purchase of reinsurance to reduce capital requirement
- Will be passing on profit as well (probably as used by insurer in last few years as per question)
- Can have Treaty reinsurance with high return commission on profitable segments
- Can be used as short term option
- Mid-term purchase can be expensive / may come with other restrictive terms like loss corridor, loss ratio caps, limited participation
- May help achieve solvency but probably not help achieve business growth

(Max 2)

Other short-term options to reduce capital requirement

- writing lower business volumes (probably as used by insurer in last few years as per question)
- Can stop writing loss making business segments / products
- Can control / stop writing long tenure business as it may be adversely impacting profits due to initial strain due to commission pay-out (which can not be deferred currently)
- Can focus only on business that can generate profit
- Can reduce / control management expenses
- Not pay dividend
- Reduce reserving strength if currently above best estimate
- May help achieve solvency but probably not help achieve business growth
- Stop writing business

(Max 2)

[10]

iv) Issues with the approach of using historical CAT events over last 20 years

- Will not represent the 1 in 200 year risk as required by EC circular
- Does not adjust for exposure change over the period (shift in business written)
- Does not allow for Increase in risk based on climate change etc.
- Does not allow for change in reinsurance arrangements over the period (proportional and non-proportional)
- Does not allow for inflation - Increase in rebuilding costs and labour costs over the period (may need to see inflation Vs Change in SI)
- Does not allow for changes in policy coverage
- Does not allow for changes in demographic shift / new industrialization / New cities developed
- Does not allow for change in building construction standards over time (and by region)
- Does not allow for improvements in Govt Public alert systems and rescue operations

[4]

v) Data items to calculate Catastrophic Risk

Data to be provided to External CAT Modeller

- By State / Pin Code level exposure for residential, commercial and industrial risks as at date
- Expected growth rate in exposure by next year (at granular level if available)
- Expected proportional reinsurance arrangements for current and prior years (so that all live policies are covered under modelling)
- Event limit, if any under proportional arrangements

Modelled output from External CAT Modeller

- AOA and AOY loss for different events with different return periods for Gross and Net exposure

- Information related to underlying Individual events – count and amount for 1 in 200 year event
- Loss cost across residential, commercial and industrial risks for same event sets
- Nearby Events which result in loss cost to 1 in 200 year AOY loss considered above
- Information on Modelled and unmodelled perils (e.g. Tsunami not covered)
- Information on modelled and unmodelled coverages (e.g. pay-out related to loss of Profit)
- Information on modelled and unmodelled policies (e.g. Coinsurance follower, Inward business, other moving risks like Marine, Motor and PA)
- Information on modelled and unmodelled policy features (e.g. auto escalation of SI, death coverage under dwellings)
- Allowance for Other features that can impact final loss like
 - Time of the event impacting alertness of people and effectiveness of rescue operations
 - Possibility to have alerts in advance and possibility to save human lives through improvements in prediction capability through advanced satellites, public information systems and
 - Effectiveness of rescue operations

Reinsurance arrangements for the year

- Retention per CAT Event (Attachment point)
- Layer wise cost
- RI Layer which covers AOY cost related to 1 in 200 year event
- Reinstatement details (Rate on Line) for all layers (which cover 1 in 200 layer event related costs)
- Max no of reinstatements
- Expected no of CAT events considered under AOY cost (input from CAT modeller) and individual loss
- Minimum deposit premium

Expected Business growth

- Expected business volumes – premiums rates and SI volumes and associated RI arrangements to achieve the budget
- Any planned deviation from existing business mix (risk categories, region etc.)
- Expected competition in the market and hence pressure on premium rate / achievement of business volumes
- Expected business volumes and Sum Insured (or Probable Maximum Loss) by pin code/State (residential, commercial, industrial type)

Any regulatory requirements

- Any common Formula at industry level – say exposure dependent
- Specified correlations – between segments
- Guidelines on recognising diversification benefits
- Confidence level – e.g. 99.5% / 99%
- Time horizon – 1 year or more
- Uncertainty with respect to future availability of CAT cover I case of longer time horizon – as generally CAT coverage will be purchased annually
- Minimum criteria for CAT modelling – say use of min 2 CAT modellers

Insurer's Risk Appetite

- Any guidance from Board on risk tolerance
- Min Confidence level criteria from credit rating agencies to maintain high Credit rating

Other

- Requirements from investors
- Comparison with capital levels maintained by Peers – so that should not look weak for other investors
- Historical capital levels
- Best practices

[10]

vi)

Motor OD Profitability as KPI

Advantages –

- easy to specify
- easy to measure (being short tail segment, not much impacted by prior years reserving strength)
- easy to understand and easy to communicate with regional underwriters

Disadvantages -

current Financial Year Loss Ratio is effect of underwriting policy in last year and current year

So, with FY LR - underwriters will be judged based on underwriting efforts during last year and current year

There will be a lag in recognising actual underwriting quality of current year due to earning effect

- So, underwriters will not get credit for improvements made in later part of the year as impact through earnings will be minimal
- Similarly, if underwriting quality has worsened over the year, it will not fully impact the underwriters' rating basis current FY LR

Short term focus – underwriting decisions can be made solely based on OD profitability.

This will not allow for

- Any cross subsidisation between Motor OD and Motor TP, if any
- e.g. if a particular segment is profitable under TP and if OD is highly subsidised then Insurer may not be present / active in that segment as OD LR will be poor for this segment
- it ignores associated acquisition costs
- it ignores associated return on capital

UW decision grid can include impact of

- acquisition costs
- investment income from float (as provided by advance premium) and balance investable portion.
- Proportion of OD and TP premium and
- expected LR based on other factors like cc/tonnage, region, vehicle age and experience
- Associated cost of capital and return on capital
- Other mandates like meeting obligatory TP quota
- Other business received apart from Motor (total life cycle value + total relationship value)
 - E.g. retail customers - other business received like health and PA, property, and association with other sister concerns and parent organisation
 - (Total relationship value derived from customer group by our group)
 - Corp customers - our share of value in other business, like Marine, property, engineering, liability, WC, GMC, GPA business
 - Separate treatment for such business where cross subsidisation is happening due to above reasons

No performance indicator can ever be 100% accurate as each will depend upon few approximations or estimates.

If the allocation method has not changed over period and business written is flat, impact of such allocation approximations/ estimation errors will be minimal and can be ignored.

Motor OD FY profitability may be overstated/ understated because of

- Expense allocation policy (assuming time sheets are not there for few common resources and further approximations with cost sharing of common resources like support functions and Property and IT hardware costs)
- Investment income allocation policy (by not having segment specific funds)
- Non-Proportional XL cost allocation (say subsidising Fire LR with high charge for CAT XL)
- Reserving related practices (case reserving of litigation claims, theft claims etc.)

Other points

- Difficult to manage expected mix at overall level (by vehicle segment, vehicle age, geography)
- Impact of business sourced through manufacture tie ups – you may not have control over premium rates, quantum of business including renewals and add-on coverages sold

- You may not have any underwriting control when business is sourced through insurer's website / mobile App – less control over inspection in case of say break-in, higher theft risk and other soft frauds.

[8]

[50 Marks]**Solution 2:****i) Reasons for starting its own captive insurance company:**

- Due to diverse risks of Oil and Gas company viz., public liability risks, Fire, Environmental liability, Engineering, Marine, Health insurance cover for employees, they cover the entire gamut of products under General Insurance Company. Hence, it is a good strategy to start Captive Insurance Company. (1)
- The company can retain the premium, which runs into '000s of crores Rs, which will otherwise be ceded to insurer. (0.5)
- This retention of premium within the group company may help to increase the strength of balance sheet of parent company. (0.5)
- The company may get savings from lesser expenses and lower profit margin compared to external insurer. (0.5)
- A captive may pass on good experience and risk management improvement savings. (0.5)
- They could get further profits by selling products to customers of the company (0.5)
- It enhances awareness to senior management of managing risk within the company. (0.5)
- Parent company will not be exposed to credit risk by external insurer. (0.5)
- Reduces exposure to the insurance cycle , thereby hardening / softening of insurance premiums in market do not affect them. This is especially important when the market conditions are hard. (1)
- Claims settlement will be quicker than that by external insurer and also there may be lesser repudiation of claims. (0.5)
- They will get direct access to reinsurance markets and expertise (0.5)
- Desired cover, terms and conditions can be negotiated with reinsurers (0.5)
- There may be tax advantages if located in certain incentivised areas, e.g. GIFT city (0.5)
- Or regulatory advantages, in particular lesser capital / reporting requirements (0.5)
- Reserves/premiums are built up as pre-tax profits from captive / parent company perspective. (0.5)
- As the parent company knows its business better, the risk exposure can be determined precisely. This is especially true for Liability, fire LoBs. (0.5)
- There may be covers unavailable in the market, they can be insured by captive insurance company. (0.5)
- Or the cover might be available but with unacceptable for the company terms and Conditions (0.5)
- The way insurance rate trends are moving, perhaps capacity/reduced premiums may not return, so the company will need to establish captive at some point. Hence, the company can start early in order to gain more experience and expertise. (1)
- If captive is permitted to insure third parties, the company may make more profits. (0.5)
- Captive may achieve higher efficiency. (0.5)
- The captive may opt for higher reinsurance; hence cost of capital may reduce. (0.5)

Disadvantages of this suggestion:

- This may not be as advantageous if the market is soft (0.5)
- Risks are being retained. (0.5)
- Risks may be inter dependent. Hence there is a possibility of accumulation of risk. (0.5)
- There will be capital requirement for captive so there is a opportunity cost of captive versus alternative better returns in other operational activities. (0.5)

- Increased volatility in parent company financial results depending on retention level post reinsurance (0.5)
- Expense of setting up and running a captive may be more than that of expense loading by insurer. (0.5)
- Specialist expertise may be required. (0.5)
- May pull management time and resource away from parent company's activities (0.5)
- Captive may not have the economies of scale as that of an insurance company. (0.5)
- Given the nature of business, high exposure to very large losses. Also, it may be impossible to get unlimited retention by reinsurer. (0.5)
- This is true for Public liability and environmental liability claims. (0.5)

[Max 15]

ii) Proposal of increasing the Excess limit:

- The XoL limit needs to be set considering the free reserves of the captive company, liquid assets of the insurer, gross written premium of the insurer apart from the nature of line of business for which XoL cover is sought. (1)
- Given the above, even Rs 100 crores excess limit for any one loss event seems to be high. Moreover, the captive insurer is in the nascent years of operation, hence it is essential to have a lower excess limit. (1)
- This is truer for liability lines of business, environmental liability LoB and other long tailed lines, where data is scanty to decide on the retention limit required. Moreover, for low frequency high severity lines like Fire, Engineering etc., it is essential to have lower excess limit. (1)
- Also, the inuring reinsurance arrangements of the insurer and their limits and terms and conditions are required to decide if the excess limit is sufficient. (0.5)
- Frequency of the loss event needs to be determined using return periods studies. Also the exposure of the business needs to be known for the severity to be determined, which will help in arriving at the excess level. (1)
- Also, 65% loading is on unadjusted average losses over last 15 years. Depending on the indexation percentage of losses (say 9% and above) considered, this premium may be lower than the average of losses of last 4-5 years, which may be favourable for increasing the retention under XoL. With high indexation, the pricing loss ratio of reinsurer also seems to be high (95%+), implying that the premium appears to be reasonable. (1.5)
- However, if the indexation percentage of claims is lower (4% or so), depending on the LoB, this premium may be higher than the average of losses of last 4-5 years. However, there will be other loadings of the reinsurer (viz., expense ratio, net commission) needs to be considered while taking the decision. (1)
- By increasing the retention limit, there will be more accumulation of risk, in case the reinsurance is a multiline XoL treaty. (0.5)
- In case the same is CAT event / terrorism risk, the insurer by itself will not have much of credible experience to compare with the premium. (0.5)
- The premium will vary depending on the insurance cycle. (0.5)
- Increasing the limit will put capital constraints, as solvency requirements may go up. (0.5)
- Increased expertise / monitoring / expenses may be required when higher limit is chosen. (0.5)
- Also, the same is dependent on the risk management strategy of the insurer. (0.5)

[Max 10]

[25 Marks]

Solution 3:

i) Reserving

- If the OD premium is going to be collected each and every time, insurance option is switched on, earning pattern will be extremely quick, hence UPR is going to be negligible for OD. (1)
- Volumes will be more variable hence, it may be difficult to arrive at a pattern for yearly / quarterly earnings. (0.5)

- It is difficult to know exactly what UPR was at a point in time and will have earned during reserving exercise. (0.5)
- It is difficult to project exposure volumes. (0.5)
- Any triangulation – even paid / incurred are unlikely to be effective given variation in earnings profile (0.5)
- Reporting delay is expected to be shorter than normal as policyholders will be reporting the claims sooner. (1)
- It is especially true if the mobile app allows electronic claims reporting (0.5)
- This should ideally lead to lower IBNR. (0.5)
- Or if the insurer questions the policyholder every time to confirm whether or not they have had any accident. (0.5)
- When multiple cars and two wheelers are covered under the same policy, the risk exposure is expected to reduce, since the same policyholder cannot drive multiple vehicles at the same time, thereby the frequency of OD may come down. However, multiple drivers are driving the vehicle, impact may not be much. (1)
- IBNER development / settlement delay on known claims likely to be similar as the same is more dependent on surveyor report / claims settlement policy etc., (0.5)
- The company may need external benchmarks for arriving at earning pattern / pilot study for the same. (0.5)
- May be difficult to estimate initial ULR. (0.5)

Pricing:

- Variable cost per policy cost is expected to increase, there by prudence will be required in expense ratio assumption. (0.5)
- The rating structure may need to have flat premium per hour depending on age of vehicle, make and model etc., (1)
- There may be more cross subsidy / selection risk between vehicles / rating factors. (1)
- Other rating factors will be the same as traditional policies. (0.5)
- Premium may get personalised down the line as more data regarding individual driver behaviour gets captured. (0.5)
- Technology cost may be higher which needs to be incorporated to policy. (0.5)
- Mostly, the product will be sold direct / online, thereby reducing the commission expenses. (1)
- The product will have different driver mix e.g. more younger drivers given internet or more city drivers given intermittent use, hence the risk characteristics will be inherently different. (1)
- Less competition in the segment. Hence insurer may charge higher margins (0.5)
- May require higher loadings for anti-selection. (1)

[Max 13]

ii)

- Short duration assets needs to be invested more, since it is expected that the claim reporting will be faster. But shorter duration assets yield will be lower, also reinvestment risk will be higher in such assets. (1)
- Higher deficit in lower years buckets will be seen if assets are picked are of longer duration. In that case macaluy duration for assets will be lower than that of liabilities leading to lack of immunization against interest rates. (1)

[2]

iii) Investment Return:

- For ALM purpose, if assets are invested for lower duration, yields in lower duration assets will be lower, compared to that of longer duration assets usually. Also, reinvestment risk will be higher in such case. (1)
- Better cash management may be required, as the receipt of income is not known in advance, hence it is better to have more free reserves in hand. (1)

- Also, the premium income may be lower compared to other standard product, which in turn may restrict the freedom to invest. (0.5)

Lapses:

- Lapses are expected to increase, as there is no incentive to renew, if the policyholder feels that a particular trip is safer (shorter distance / lower traffic route), which may lead to anti selection. (1)
- Increased interaction with policyholder through touch points of insurer may increase retention. (0.5)
- If not many insurers in the market offer such product, existing policyholder will stick with the insurer and up selling opportunities may increase. (0.5)

Expenses and Commission:

- Cost of device and other headers viz., maintenance cost of the device / app will increase, which will increase the cost to customer. Insurer may take up a share of this expense. (1)
- Data tracking and administration expenses will increase. (0.5)
- Depending on the target group, which is mostly expected to be young drivers and city based, same may be sold online / direct, which in turn will reduce commissions. (1)

Customer behaviour:

- Anti selection may increase depending on the type of trip taken. OD cover may be opted only for riskier trips. (0.5)
- Customers may behave properly assuming that they are being tracked. (0.5)
- Multiple drivers using the same vehicle may reduce, as the policyholder may sense that the insurance company will track them. (1)

Capital:

- Free capital may be required for countering uncertainties in claim pattern, earning pattern etc., (0.5)
- If claim experience is better, same may reduce the capital requirement. (0.5)
- Given that IBNR may be low, same will reduce capital required. (0.5)

**[Max 10]
[25 Marks]**
